

Algebra II with Trigonometry

Chapter 7.1

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4564659_alg-2trig-h-chp-7-1-note.pdf](#)

1. **Quotient Identity** Simplify mentally: $\frac{\sin x}{\cos x}$
2. **Reciprocal Identity** Simplify mentally: $\sin x \cdot \csc x$
3. **Pythagorean Identity** Simplify mentally: $1 + \tan^2 x$
4. **Even-Odd Check** Rewrite without a negative angle: $\cos(-\theta)$
5. **Odd Function Check** Rewrite without a negative angle: $\sin(-\theta)$
6. **Cofunction Identity** Simplify: $\sin\left(\frac{\pi}{2} - x\right)$
7. **Cofunction Identity** Simplify: $\tan\left(\frac{\pi}{2} - x\right)$
8. **Convert to Sine and Cosine** Rewrite using only sine and cosine: $\cot x$
9. **Quick Simplification** Simplify: $\cos x \cdot \sec x$
10. **Difference of Squares with Trig** Simplify: $\sin^4 \theta - \cos^4 \theta$